

Prime Conductors versus a Fixed Atom: Recurrence and Period-Corridor Separation

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Abstract

A fixed nonzero rational atom cannot be represented in lowest terms with infinitely many distinct prime conductors q_X . The variable q_X branch also lies outside the source-locked polylogarithmic exact-period corridor; the conductor-one branch remains possible but has no occurrence or packet schedule theorem. The classification is `CONSISTENCY_BARRIER_L1` and the verdict is `PROVED_NONZERO_FIXED_RATIONAL_ATOM_CANNOT_RECUR_ACROSS_UNBOUNDED_PRIME_CONDUCTORS`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains `STOP-SCOPED` [2].

2 Reduced-denominator rigidity

The phase-conductor ledger gives a reduced conductor in $\{1, q_X\}$, with the nonconstant branch carried by unbounded prime values q_X [3, 4].

Theorem 1 (Fixed-atom consistency). *Let $\alpha_\star = r_0/R_0 \pmod{1}$ be reduced and nonzero. If $\alpha_\star = r_X/q_X \pmod{1}$ is also reduced for infinitely many distinct primes q_X , then $R_0 = q_X$ for infinitely many distinct primes, a contradiction. Hence a nonzero fixed rational atom cannot recur through the variable-prime-conductor branch.*

Proof. Reduced representatives of the same rational point on $\mathbb{R}/\mathbb{Z}$ have the same positive denominator. Thus $R_0 = q_X$ at every such scale. $\square$

3 Exact-period corridor

The periodic approximation theorem requires

$$q_{\text{prog}} R \leq (\log X)^{\eta_0}.$$

On the variable branch $R = q_X = X^{267/400+o(1)}$, this fails for all sufficiently large X , even before the additional factor $q_{\text{prog}} \geq 1$ is charged [1]. On the conductor-one branch $R = 1$, only the variable-conductor obstruction disappears. A valid invocation still requires

$$q_{\text{prog}} \leq (\log X)^{\eta_0}, \quad N \in [\sqrt{X}, X] \setminus E_X^\star,$$

together with every other native good-scale condition. The repository supplies neither those schedule-specific admissibility facts, a source-locked occurrence of a named $\alpha_\star$, nor an exact production packet schedule on all required scales.

4 Scope

This is a consistency theorem, not fixed-atom cancellation. It prunes one possible recurrence mechanism while leaving the conductor-one and genuinely nonperiodic pointwise routes open.

5 Loss, level and scope ledger

The smallest missing literal theorem is

`SOURCE_LOCKED_CONDUCTOR_ONE_OCCURRENCE_PACKET_SCHEDULE_AND_RANGE_ADMISSIBILITY.`

The next route is `FIXED_ATOM_OCCURRENCE_EDGE`. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

6 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Liang Wang. Literal weight periodic approximation, 2026.
- [2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.
- [3] Liang Wang. Literal low-window affine export, 2026.
- [4] Liang Wang. Exact content resonance ledger, 2026.